

AutoDock Vina 1.2.7 (32-type XS)



extends

LKina

113 types · metal-aware · covalent-reactive

Metal-coordination docking



TZ / SQ / MH / JT pseudoatoms
BVS oxidation-state inference
4JC Zn-NA at 2.13 Å
(Vina 2.23 Å, no coord)

Inline AutoGrid 4.2 — no external autogrid4
108 AG4-format maps incl. TZ/SQ/MH/JT

Covalent-reactive docking



NAC detection ($d < 3.0$ Å, angle)
P1-P4 · C3 two-step · 6 presets
end-to-end, NAC discriminates
covalent vs non-covalent

Backward compatible — Vina 1.2.7 energy identical
1HVR -14.54 · 3PTB -6.202 kcal/mol (mode-1)

Metal-mode coverage: Vina 1.2.7 34/110 → LKina 110/110